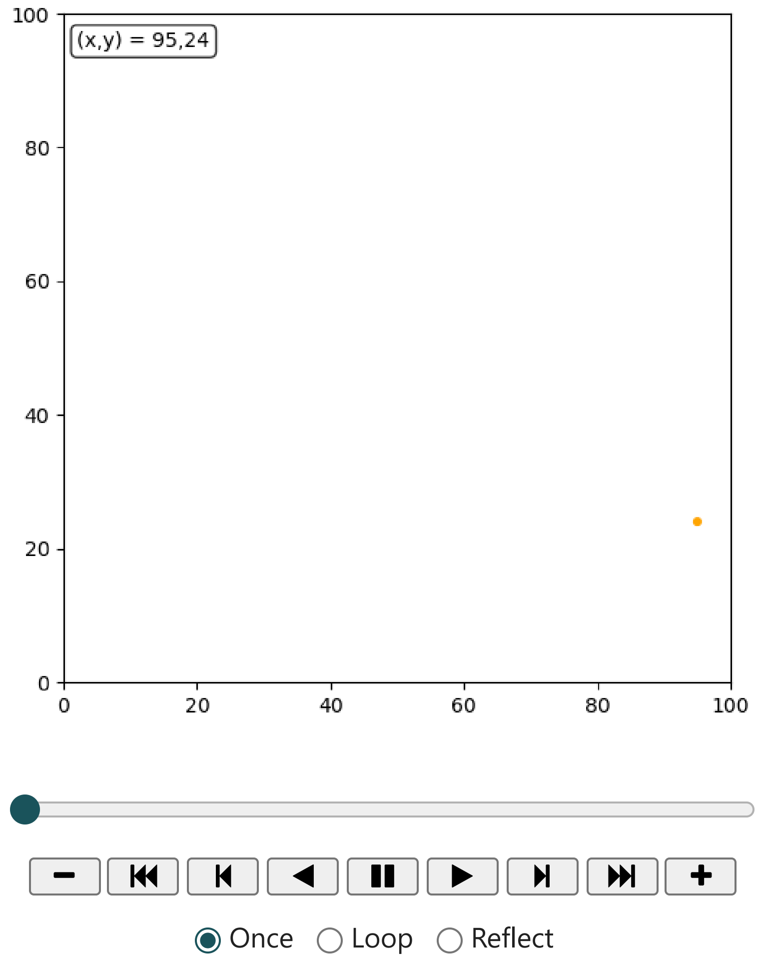
In [3]:
```python
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation
from itertools import count
from random import randint as randint
from random import uniform as uniform

%matplotlib inline

x0 = np.random.randint(0,100)
y0 = np.random.randint(0,100)

def move_particle():
    d = uniform(0, 1)
    dx, dy = 0, 0

    if (d <= 0.25):
        dy = 2                          # North
    elif (d > 0.25 and d <= 0.5):
        dx = 2                          # East
    elif (d > 0.5 and d <= 0.75):
        dy = -2                         # South
    else:
        dx = -2                         # West
    return dx, dy

def get_pos():
    x, y = x0, y0
    t = 0
    while t < 10000:
        dx, dy = move_particle()
        t += 1
        x += dx
        y += dy

        yield x, y

def init():
    ax.set_xlim(0, 100)
    ax.set_ylim(0, 100)
    line.set_data([], [])
    ball.set_center((x0, y0))
    height_text.set_text(f'(x,y) = {x0},{y0}')
    return line, ball, height_text

def animate(pos):
    x, y = pos
    xdata.append(x)
    ydata.append(y)

    if len(xdata) > 1000:
        xdata.pop(0)
        ydata.pop(0)
```

```
    line.set_data(xdata, ydata)
    ball.set_center((x, y))
    height_text.set_text(f'(x,y) = {x},{y}')

    current_xlim = ax.get_xlim()
    current_ylim = ax.get_ylim()

    if x < current_xlim[0] or x > current_xlim[1]:
        ax.set_xlim(min(x-10, current_xlim[0]), max(x+10, current_xlim[1]))
    if y < current_ylim[0] or y > current_ylim[1]:
        ax.set_ylim(min(y-10, current_ylim[0]), max(y+10, current_ylim[1]))

    return line, ball, height_text

fig, ax = plt.subplots(figsize=(8, 6))
ax.set_aspect('equal')

line, = ax.plot([], [], lw=2, alpha=0.7)
ball = plt.Circle((x0, y0), radius=0.5, color='orange')
ax.add_patch(ball)
xdata, ydata = [], []
height_text = ax.text(0.02, 0.95, f'(x,y) = {x0},{y0}', transform=ax.transAxes,
                     bbox=dict(boxstyle="round,pad=0.3", facecolor="white", alpha=0

interval = 100
ani = animation.FuncAnimation(fig, animate, get_pos, init_func=init,
                            interval=interval, repeat=False, blit=False,
                            save_count=1000, cache_frame_data=False)

# For Jupyter notebook - display as HTML animation
from IPython.display import HTML
HTML(ani.to_jshtml())
```

Out[3]:



In [ ]: